

LINEAR REGRESSION

Simple Linear Regression Equation

$$y = mx + c$$

or for prediction,

$$\hat{y} = mx + c$$

where:

- y = actual output
- $\hat{y}$ = predicted output
- x = input variable
- m = slope (regression coefficient)
- c = intercept (constant term)

1 Problem Formulation

Input space:

The input space is defined as

$$\mathcal{X} = \mathbb{R}^p$$

Each observation is a feature vector

$$x = (x_1, x_2, \dots, x_p)^T$$

where the superscript T denotes transpose, meaning the feature vector is represented as a column vector.

Output space:

$$\mathcal{Y} = \mathbb{R}$$

The target variable is continuous and real-valued.

Data distribution:

Assume the training data are independently and identically distributed (i.i.d.) samples drawn from an unknown joint probability distribution

$$(x, y) \sim P(X, Y)$$

The dataset is

$$D = \{(x_i, y_i)\}_{i=1}^n$$

Learning objective (Expected Risk Minimization):

The goal is to find a function f that minimizes the expected risk

$$R(f) = \mathbb{E}_{(X, Y)} [(Y - f(X))^2]$$

Since the true distribution $P(X, Y)$ is unknown, we minimize the empirical risk computed from the sample.



2 Model Specification

Hypothesis function:

Linear regression assumes a linear relationship between input and output:

$$f_w(x) = w^T x + b$$

where

- $w \in \mathbb{R}^p$ is the parameter vector
- w^T denotes transpose of w
- $b \in \mathbb{R}$ is the bias term

In augmented form (including bias inside parameters):

$$f_w(x) = w^T x$$

Parameter space:

$$w \in \mathbb{R}^p$$

Structural assumptions:

1. Linearity in parameters
2. Additive noise model
3. No perfect multicollinearity
4. Independent errors

Model assumption:

$$Y = w^T X + \epsilon$$

3 Loss Function

Explicit mathematical form:

Squared loss function is

$$\ell(y, \hat{y}) = (y - \hat{y})^2$$

Why this loss? (Statistical reasoning):

If noise follows a Gaussian distribution

$$\epsilon \sim \mathcal{N}(0, \sigma^2)$$

then maximizing likelihood is equivalent to minimizing squared error. Thus squared loss corresponds to the Gaussian noise assumption.

Convexity properties:

The squared loss is convex because its Hessian matrix is

$$\nabla^2 (y - w^T x)^2 = 2xx^T$$

Since xx^T is positive semi-definite, the loss function is convex and guarantees a global minimum.

4 Objective Function

Empirical Risk expression:

$$J(w) = \frac{1}{n} \sum_{i=1}^n (y_i - w^T x_i)^2$$

Matrix form:

$$J(w) = \frac{1}{n} (y - Xw)^T (y - Xw)$$

where

- $X \in \mathbb{R}^{n \times p}$
- X^T denotes transpose of X
- $y \in \mathbb{R}^n$

Regularized formulation (Ridge regression):

$$J(w) = \frac{1}{n} (y - Xw)^T (y - Xw) + \lambda w^T w$$

where $\lambda > 0$ controls model complexity.

5 Optimization Method

Closed-form solution:

Taking derivative and setting to zero gives the normal equation:

$$w^* = (X^T X)^{-1} X^T y$$

provided $X^T X$ is invertible.

Gradient expression:

$$\nabla J(w) = -\frac{2}{n} X^T (y - Xw)$$

Iterative optimization:

Using Gradient Descent:

$$w_{t+1} = w_t - \eta \nabla J(w_t)$$

where η is the learning rate.

Convergence guarantees:

Since the objective function is convex and quadratic, gradient-based methods converge to the global minimum.

Computational complexity:

Closed-form solution:

$$O(p^3)$$

Gradient descent:

$$O(np) \text{ per iteration}$$

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6 Statistical Interpretation

Maximum Likelihood Estimation (MLE) connection:

Assume

$$Y = Xw + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2 I)$$

Then the Ordinary Least Squares estimator equals the Maximum Likelihood Estimator.

Maximum A Posteriori (MAP) interpretation:

If prior is

$$w \sim \mathcal{N}(0, \tau^2 I)$$

then Ridge regression corresponds to MAP estimation.

Noise model assumption:

1. Zero mean
2. Constant variance (homoscedasticity)
3. Independent errors

Probabilistic meaning of outputs:

$$Y | X \sim \mathcal{N}(w^T X, \sigma^2)$$

Predictions represent the conditional mean.

7 Regularization & Generalization

Bias-variance tradeoff:

- No regularization → low bias, high variance
- Ridge regression → higher bias, lower variance

Overfitting behavior:

Occurs when number of features is large relative to samples or when multicollinearity exists.

Capacity control mechanism:

Controlled by parameter norm $w^T w$ and regularization parameter λ .

8 Theoretical Properties

Convexity and global optimality:

The objective function is a convex quadratic function. If $X^T X$ is full rank, the solution is unique and globally optimal.

Consistency:

If the model is correctly specified and sample size increases to infinity, the estimator converges to the true parameter.

Stability considerations:

Linear regression is sensitive to the condition number of $X^T X$. Poor conditioning leads to unstable estimates.

9 Computational Complexity

Let

n = number of training samples (observations)

p = number of input features (dimension of each feature vector)

Thus, the design matrix

$$X \in \mathbb{R}^{n \times p}$$

contains n rows (samples) and p columns (features).

Training time complexity:

For the closed-form solution

$$w^* = (X^T X)^{-1} X^T y$$

the dominant cost comes from computing and inverting the matrix $X^T X$, where

$$X^T X \in \mathbb{R}^{p \times p}$$

Matrix multiplication $X^T X$ requires

$$O(np^2)$$

Matrix inversion requires

$$O(p^3)$$

Thus overall training complexity (closed-form) is

$$O(np^2 + p^3)$$

When p is large, the $O(p^3)$ term dominates.

For Gradient Descent, each iteration requires computing

$$X^T (y - Xw)$$

which costs

$$O(np)$$

If the algorithm runs for T iterations, total complexity is

$$O(npT)$$

Inference time complexity:

For predicting one new sample $x \in \mathbb{R}^p$, we compute

$$\hat{y} = w^T x$$

which requires

$$O(p)$$

operations.

Memory complexity:

Storing the design matrix X requires

$$O(np)$$

memory.

Storing the parameter vector $w \in \mathbb{R}^p$ requires

$$O(p)$$

memory.

Therefore, total memory requirement is dominated by

$$O(np)$$

10 Limitations

When it fails:

- Non-linear relationships
- Presence of strong outliers
- High-dimensional case where $p \gg n$

Assumption violations:

- Heteroscedasticity
- Correlated errors
- Multicollinearity

Sensitivity issues:

- Sensitive to outliers due to squared loss
- Sensitive to feature scaling (for gradient methods)
- Unstable when $X^T X$ is ill-conditioned